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BENCHMARKING MODERN AI CHATBOTS: A MULTIDIMENSIONAL EVALUATION OF CHATGPT, GEMINI, AND COPILOT

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ABSTRACT

Artificial Intelligence (AI) chatbots have gained popularity for tasks such as information retrieval and academic assistance, yet their effectiveness in regional languages remains less explored compared to their optimization for global languages like English. This study investigates user perceptions of AI chatbot performance in regional languages, utilizing a survey that involved diverse user groups—students, teachers, and professionals. It evaluates preferences among prominent chatbots, including ChatGPT and Gemini, analyzing various performance metrics such as language understanding, response quality, grammatical accuracy, contextual awareness, response time, and overall usability. Statistical analyses, specifically the Chi-Square Test of Independence, One-Way ANOVA, and the Friedman Test, were employed to discern significant patterns within user preferences and evaluation scores. Visualization methods, including bar charts, heatmaps, box plots, and radar charts, were utilized to illustrate comparative performance insights. Results indicate that user role significantly affects chatbot preference, while performance ratings tend to be consistent across different age groups. The findings emphasize user perceptions of AI chatbots in regional languages and suggest potential improvements for multilingual AI systems, enhancing both accessibility and user experience.

Keywords: AI Chatbots, Multilingual NLP, ChatGPT, Gemini, Copilot, User Perception, Conversational AI, Regional Languages.

I. INTRODUCTION

Artificial Intelligence has significantly transformed the way users interact with digital systems, particularly through the development of AI chatbots capable of understanding and generating natural language. Modern conversational systems such as ChatGPT, Gemini, and Microsoft Copilot are increasingly used in areas like education, customer support, and information retrieval because they can provide quick and relevant responses to user queries. Despite these advancements, most chatbot systems are primarily optimized for global languages, especially English. This creates challenges in multilingual regions such as India, where a large portion of users prefer to communicate in regional languages.

The limited optimization of AI chatbots for regional languages raises important questions regarding their effectiveness in multilingual communication. Issues related to language understanding, grammatical accuracy, contextual interpretation, and clarity of responses become particularly significant when users interact with these systems in non-English languages. Therefore, evaluating how well modern AI chatbots perform in such linguistic environments is an important research objective.

This study examines user perceptions of AI chatbot performance in regional language interactions through a structured survey involving participants from different demographic groups. The evaluation focuses on key factors such as language comprehension, response accuracy, contextual relevance, response quality, and overall usability. To analyze the collected data, statistical techniques including the Chi-Square Test of Independence, One-Way ANOVA, and the Friedman Test were applied. In addition, visualization methods were used to present comparative insights into chatbot performance.

The findings of this research contribute to a better understanding of how users perceive AI chatbot capabilities when communicating in regional languages. The results also highlight potential areas for improvement in multilingual conversational systems, helping developers design AI tools that are more accessible, inclusive, and effective for diverse linguistic communities.

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II. LITERATURE REVIEW

Recent advancements in generative artificial intelligence (AI) and large language models (LLMs) have significantly influenced various domains, including education, healthcare, programming, and cybersecurity. Early research highlights the growing integration of AI chatbots such as ChatGPT, Gemini, and Microsoft Copilot in academic and professional environments. These models demonstrate strong capabilities in natural language processing, reasoning, and automated content generation, enabling new approaches to digital learning and information access [1].

In the healthcare domain, several studies have evaluated the performance and reliability of LLMs in providing medical information. For example, research on keratoconus-related queries assessed the ability of multiple chatbots, including ChatGPT-3.5, ChatGPT-4, Gemini, and Copilot, to deliver accurate and readable responses for patient education. The findings indicated that while LLMs can provide informative responses, their reliability and readability vary depending on the model and the complexity of the medical questions [2]. Similarly, another study examined the capability of AI systems to answer nursing examination questions. The results showed that AI models such as ChatGPT-3.5, Gemini, and Copilot achieved notable accuracy levels in nursing assessments, with Copilot achieving the highest score in certain tasks [3].

Further investigations have explored the performance of LLM chatbots in medical education and neuroscience. Comparative analyses between AI chatbots and students demonstrated that advanced models such as Claude and GPT-4 can achieve accuracy levels comparable to or even exceeding those of students in specific neuroscience topics, although performance varies across subject areas [4]. These findings highlight the potential of LLMs as supplementary learning tools in medical education.

Alongside educational benefits, researchers have also raised concerns regarding cybersecurity risks associated with generative AI technologies. Studies indicate that AI chatbots can be exploited for malicious purposes such as phishing attacks, social engineering, and the generation of harmful code. Consequently, the development of advanced security mechanisms and ethical guidelines is necessary to mitigate these threats while ensuring responsible AI deployment [5].

In the field of language education, generative AI tools have also been studied for their impact on writing instruction. Comparative research evaluating ChatGPT-4, Gemini, and Copilot found that these tools can effectively assist students in brainstorming, outlining, and revising essays. However, their effectiveness varies depending on the writing stage and the quality of generated feedback [6].

Beyond education, LLMs are increasingly used in software development. Studies comparing AI-assisted programming tools such as ChatGPT, Gemini, AlphaCode, and GitHub Copilot show that these systems can generate functional code and assist programmers with debugging and problem solving. Nevertheless, issues related to reliability, ethical concerns, and human oversight remain critical challenges in practical implementation [7].

AI chatbots are also transforming academic library services by providing automated assistance and improving access to information resources. Research suggests that chatbot integration can reduce routine workloads for librarians and enhance service availability, especially in institutions with limited staffing or resources [8].

In the healthcare communication domain, researchers have evaluated the readability and appropriateness of AI-generated responses related to refractive surgery. Findings indicate that LLM chatbots can provide understandable and structured explanations for patient queries, although their accuracy and medical reliability still require careful validation [9].

Research Gap (Short Section)

Although numerous studies have evaluated the capabilities of large language models such as ChatGPT, Gemini, and Microsoft Copilot across domains including healthcare, education, and programming, most existing research primarily focuses on task-specific performance, accuracy of responses, or domain-based applications [2]–[7]. Some studies have also investigated the use of generative AI tools in educational support and language learning environments [1], [6], while others have examined their implications in specialized contexts such as medical information delivery and cybersecurity [5], [9].

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However, limited research has systematically examined the comparative performance of these AI chatbots in multilingual or regional language environments, particularly from the perspective of user perception and real-world usability. In multilingual countries such as India, where users frequently interact with digital systems in regional languages, the effectiveness of AI chatbots in understanding and generating contextually appropriate responses remains insufficiently explored. Furthermore, prior studies rarely combine user surveys, human evaluation, response time analysis, and error analysis within a single evaluation framework.

Therefore, this study addresses this gap by conducting a multidimensional comparative evaluation of ChatGPT, Gemini, and Microsoft Copilot, focusing specifically on regional language interaction (Marathi, Hindi, and English) and incorporating both technical performance metrics and user perception analysis.

III. METHODOLOGY OF STUDY

A. Evaluation Methodology

Four complementary metrics were used to assess the effectiveness of three AI conversational systems: ChatGPT, Google Gemini, and Microsoft Copilot.

1. User Survey Analysis: Students' and instructors' opinions about chatbot usability and regional language performance were gathered through a Google Form survey.
2. Human Evaluation: After comparing the three chatbots' anonymized responses in Marathi, Hindi, and English, participants chose their favorite response.
3. Response Time Analysis: Complex reasoning prompts were used to gauge response latency.
4. Error Type Analysis: To evaluate correctness and conceptual accuracy, AI-generated responses were contrasted with manually resolved reference answers.

This multi-metric evaluation framework made it possible to evaluate response accuracy, computational efficiency, linguistic proficiency, and user perception.

B. Google Form Survey Analysis

1) Participant Demographics

Role	Percentage
Students	57%
Teachers/Faculty	39%
Others	4%

The dominance of student participants reflects the primary user group of AI-based academic assistance tools.

2) Preferred Chatbot for Regional Languages

Chatbot	Responses	Percentage
ChatGPT	22	78.60%
Google Gemini	6	21.40%
Microsoft Copilot	0	0%

ChatGPT received almost four times as many votes as Gemini, indicating a strong preference, according to survey results. Particularly in regional language interactions, respondents claimed that its answers were more contextually accurate, natural, and clear.

3) Perceived Performance

Participants assessed the chatbot's performance in terms of accuracy, response time, naturalness of language, grammatical correctness, and contextual continuity. The majority of participants expressed a willingness to use AI chatbots in academic and multilingual communication contexts, and overall responses showed favorable opinions of these systems.

C. Human Evaluation Analysis

Chatbot identities were concealed during evaluation to reduce bias. Participants compared responses in Marathi, Hindi, and English, with findings revealing a stronger preference for English responses. Growing acceptance of Marathi responses was noted, especially among students, while Hindi responses received moderate preference. Participants favored responses that exhibited clarity, logical justification, organized presentation, and grammatical correctness, indicating that human preference is influenced by linguistic fluency and structured reasoning.

D. Response Time Analysis

Five complex prompts involving logical reasoning, algorithm design, probability, system architecture, and formal language theory were used to measure response latency.

Response Time Comparison (seconds)

Prompt	ChatGPT	Gemini	Copilot
Q1	29.87	22.51	61.06
Q2	39.85	12.14	33.21
Q3	24.66	17.18	52.01
Q4	47.9	18.14	34.74
Q5	35.22	13.18	61.34

Average Response Time

According to the results, Microsoft Copilot had the longest response times, ChatGPT had moderate latency, and Google Gemini generated responses the fastest. But quicker reaction times don't always translate into higher-quality reasoning.

Model	Average Time (s)
Gemini	16.63
ChatGPT	35.5
Copilot	48.47

E. Error Type Analysis

In a number of academic fields, error analysis contrasted AI responses with manually resolved reference solutions:

- Statistics
- Java Programming
- Python Programming
- C Programming
- Database Management Systems (DBMS)
- Data Structures

Key Findings

With differing levels of algorithm efficiency and explanation quality, the models generated syntactically correct code structures. While ChatGPT excelled in structured logic and unambiguous variable name, Gemini produced optimized algorithms. Sometimes syntactic problems were caused by Copilot. All models provided accurate explanations of the LIFO principle and operations such as push and pop for conceptual questions, with Gemini

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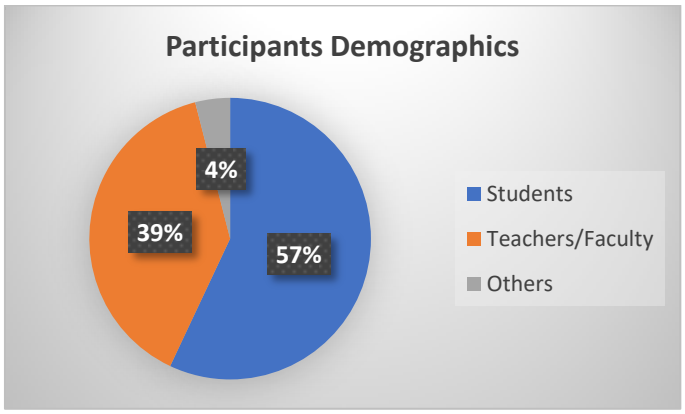
supplying comprehensive descriptions and ChatGPT providing applicable real-world examples. All models consistently produced precise SQL queries in database searches to identify students with scores above average.

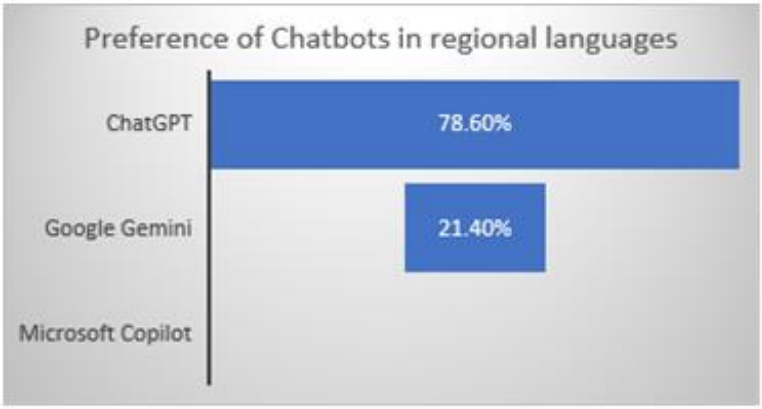
F. Comparative Performance Summary

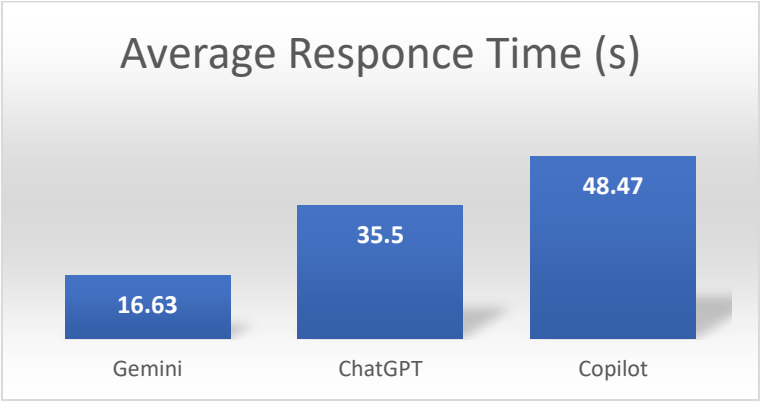
Metric	Best Performing Model
User Preference	ChatGPT
Human Evaluation	ChatGPT
Response Time	Gemini
Error Accuracy	ChatGPT / Gemini

These findings show that ChatGPT performs better in terms of user satisfaction and explanation quality, while Gemini excels in response speed.

G. Graphical Representation







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H. Statistical Hypothesis Testing

To further analyze the collected survey data, statistical hypothesis testing techniques were applied. These tests help determine whether the observed differences in user responses are statistically significant or occur due to random variation. Three statistical methods were used in this study: Chi-Square Test of Independence, One-Way ANOVA, and the Friedman Test.

A significance level of $\alpha = 0.05$ was used for all hypothesis tests.

H.1 Chi-Square Test of Independence

The Chi-Square Test of Independence was conducted to determine whether there is a significant relationship between the user role and the preferred AI chatbot for regional language interaction. This test is appropriate because both variables are categorical in nature. It helps identify whether different user groups demonstrate different preferences toward AI chatbots.

Hypotheses

H_0 (Null Hypothesis):

User role and chatbot preference are independent.

H_1 (Alternative Hypothesis):

User role and chatbot preference are associated.

Observed Frequency Table

User Role	ChatGPT	Gemini
Farmer	0	1
Student	15	1
Teacher/Faculty	7	4

Test Results

Statistic	Value
Chi-Square Statistic	**7.31**
Degrees of Freedom	**2**
p-value	**0.0258**

Interpretation

Since the p-value (0.0258) is less than the significance level of 0.05, the null hypothesis is rejected.

This indicates that chatbot preference is significantly associated with user role. In other words, different user groups (students, teachers, etc.) demonstrate different preferences when selecting AI chatbots for regional language interactions.

H.2 One-Way ANOVA Test

A One-Way Analysis of Variance (ANOVA) test was performed to determine whether there are statistically significant differences in the overall chatbot performance rating across different age groups.

ANOVA is used when comparing the mean values of more than two groups.

Hypotheses

H_0 (Null Hypothesis):

The mean chatbot performance rating is the same across all age groups.

H_1 (Alternative Hypothesis):

The mean chatbot performance rating is different for at least one age group.

Test Results

Statistic	Value
F-Statistic	**0.71**
p-value	**0.553**

Interpretation

The p-value (0.553) is greater than 0.05, therefore the null hypothesis cannot be rejected.

This indicates that there is no statistically significant difference in overall chatbot performance ratings across different age groups. Users from different age categories tend to evaluate chatbot performance similarly.

H.3 Friedman Test for Human Evaluation

The Friedman Test was used to compare multiple human evaluation metrics collected through Likert-scale survey questions. Since Likert scale data is ordinal and may not follow a normal distribution, the Friedman Test is a suitable non-parametric alternative to repeated-measures ANOVA.

The evaluation factors included:

- Language understanding
- Grammatical correctness
- Context maintenance
- Naturalness of responses
- Information accuracy
- Response time satisfaction
- User comfort during interaction

Hypotheses

H_0 (Null Hypothesis):

There is no significant difference among the evaluation metrics.

H_1 (Alternative Hypothesis):

At least one evaluation metric differs significantly.

Test Results

Statistic	Value
Friedman Chi-Square	**5.48**
p-value	**0.483**

Interpretation

The p-value (0.483) is greater than the significance level of 0.05, therefore the null hypothesis cannot be rejected.

This suggests that user ratings across the different evaluation criteria are relatively consistent, indicating that participants perceive chatbot performance similarly across multiple aspects of interaction.

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H.4 Summary of Statistical Findings

Test	Purpose	Result
Chi-Square Test	Relationship between user role and chatbot preference	Significant relationship found
ANOVA Test	Differences in ratings across age groups	No significant difference
Friedman Test	Comparison of multiple human evaluation metrics	Ratings are consistent

These statistical analyses provide additional validation for the survey findings and help establish whether observed differences in chatbot performance and user perception are statistically meaningful.

IV. RESULTS AND ANALYSIS

The results of the study highlight notable differences in the performance of modern AI chatbots across multiple evaluation dimensions. Survey responses indicated that ChatGPT received the highest overall user preference, particularly for its clear explanations, contextual understanding, and natural language fluency. In contrast, Google Gemini demonstrated the fastest response times, making it suitable for situations where quick interaction is important. Microsoft Copilot produced generally accurate outputs but showed comparatively higher response latency in several test prompts.

Human evaluation further confirmed that users preferred responses that were well-structured, logically explained, and grammatically clear. Participants also expressed greater comfort when chatbot responses closely resembled natural human language. The error analysis revealed that while all three models could generate correct conceptual explanations and programming solutions, the quality of explanations and algorithm clarity varied between systems.

Overall, the findings suggest that response quality, clarity, and contextual reasoning play a more significant role in user satisfaction than response speed alone. These results highlight the importance of balancing computational efficiency with linguistic and reasoning capabilities when evaluating conversational AI systems.

V. DISCUSSION

The comparative analysis demonstrates that evaluating AI chatbots requires a multidimensional approach that considers both technical performance and user perception. While Google Gemini delivered faster responses, ChatGPT achieved higher user preference due to its stronger contextual reasoning and more natural language explanations.

These findings suggest that in educational and knowledge-based environments, users value clarity and explanation quality more than response speed. The results also emphasize the growing importance of regional language support, as many users prefer interacting with AI systems in their native languages for better comprehension and accessibility.

VI. LIMITATIONS OF THE STUDY

Despite providing valuable insights into chatbot performance, this study has several limitations. First, the analysis is based on a relatively small survey sample, which may restrict the generalizability of the findings. A larger and more diverse participant pool could provide broader insights into user perceptions across different demographic groups.

Second, the human evaluation process involves subjective judgment, as participants evaluated responses based on personal interpretation of clarity, usefulness, and correctness. Although chatbot identities were anonymized to minimize bias, individual preferences and familiarity with AI tools may still influence evaluations.

Third, the study focuses on a limited set of languages—Marathi, Hindi, and English—and a restricted number of test prompts for response time and error analysis. Expanding the study to include additional regional languages and a wider range of tasks could offer a more comprehensive understanding of chatbot capabilities.

Finally, AI chatbot technologies are continuously evolving, with frequent updates and improvements. Consequently, the performance results reported in this study represent the systems at the time of evaluation and may change as newer model versions become available.

VII. CONCLUSION

This study presented a multidimensional evaluation of three widely used AI conversational systems: ChatGPT, Google Gemini, and Microsoft Copilot. Using survey analysis, human evaluation, response time measurement, and error analysis, the study examined both the technical performance and user perception of these chatbots.

The results indicate that ChatGPT achieved the highest user preference and evaluation scores, primarily due to its clear explanations and contextual understanding. Google Gemini demonstrated the fastest response times, while Microsoft Copilot showed comparatively slower response performance. Overall, users preferred chatbot responses that were logically structured, easy to understand, and linguistically natural.

These findings suggest that modern AI chatbots are increasingly effective tools for educational support, programming assistance, and multilingual communication, although improvements are still needed to enhance their performance in regional language interactions.

VIII. FUTURE SCOPE

Future research can expand this study by including larger and more diverse participant groups to obtain more representative insights into user perceptions of AI chatbots. Additionally, evaluating chatbot performance across a broader range of regional languages and real-world tasks would provide a deeper understanding of multilingual conversational AI capabilities.

Further studies may also explore longer conversational interactions, domain-specific applications, and real-time user feedback mechanisms to better assess chatbot reliability in practical scenarios. Improving regional language fluency, reducing hallucination errors, and enhancing contextual understanding will be key areas of development for future AI conversational systems.

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